

COURSE PROJECT

SMART TOURIST SAFETY AND TRACKING SYSTEM

School of Computer Science Engineering and Information Systems (SCORE)

Winter Semester 2025-2026

ISWE207L-OBJECT ORIENTED ANALYSIS AND DESIGN

SLOT: D1+TD1

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1. Objectives

The main objective of this project is to design a Smart Tourist Safety and Tracking System that helps improve the safety of tourists during travel. The system mainly focuses on tracking location, detecting risky situations, and sending alerts quickly.

The specific objectives are:

- To design a system that can track tourist location in real-time using GPS technology.
- To reduce response time during emergency situations by automatically sending alerts to admin and family members.
- To develop a simple system structure that includes modules like tracking, alert system, and monitoring.
- To detect abnormal situations such as unusual movement or no movement for a long time.
- To provide a user-friendly interface for tourists and admin to use the system easily.
- To improve communication between tourist, admin, and family members during critical situations.

This system mainly aims to provide safety, quick response, and better coordination.

2. Literature Review

Tourist safety is an important issue, especially in places where people travel frequently. Many systems are used today, but most of them are basic and not fully effective.

In many places, safety depends on emergency numbers, local help, or manual tracking. Some mobile apps provide location sharing, but they depend on the user to send alerts manually.

In real situations, tourists may not be able to send alerts because of panic, injury, or network problems. This creates delay and increases risk.

In some countries, advanced systems are used with GPS tracking and monitoring. These systems can track location and send alerts, but they are expensive and complex.

There are also systems that use wearable devices and sensors to monitor user activity. But these systems require special devices and are not easy to use in all situations.

Even with these technologies, many problems still exist. Most systems focus only on tracking and not on automatic alert generation.

This project tries to solve these problems by designing a simple, cost-effective, and easy-to-use system that combines tracking, alert, and monitoring in one platform.

3. Methodology

The methodology of this project focuses on understanding the problem and designing a system using OOAD concepts.

The approach consists of three main stages: system analysis, design, and workflow development.

First, the problem is analyzed and the main components are identified such as Tourist, Admin, Alert, and Device.

Then, the system is designed using UML diagrams to represent structure and behavior.

Finally, the workflow of the system is created to show how the system works step by step.

3.1 Data Collection

Data for this project is collected from different sources.

Some information is taken from online articles related to tourist safety. News reports about missing tourists and accidents are also studied.

We also observed how current systems work and what problems they have. In addition, class notes and study materials are used.

The collected information helped us to understand real-world problems and design a better system.

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3.2 Tools and Technologies

The system is designed using modern web technologies and basic software tools.

The frontend is planned using React.js to create a simple and interactive user interface. It helps users like tourists and admin to easily use the system.

The backend is designed using Node.js and Express.js. It handles system logic, processes data, and manages communication between different components.

MongoDB is used as the database to store data like user details, location history, and alert records.

GPS technology is used to track the real-time location of tourists. A basic AI concept is used to detect abnormal situations and generate alerts automatically.

UML diagrams are created using PlantUML to design the system structure. Version control is maintained using GitHub to manage project files.

3.3 System Workflow

The system starts when the tourist logs into the application. After login, the tourist can start the trip.

Once the trip starts, the GPS tracking system begins to collect location data continuously.

The system monitors the movement of the tourist and checks for any abnormal conditions.

If everything is normal, the tracking continues in the background.

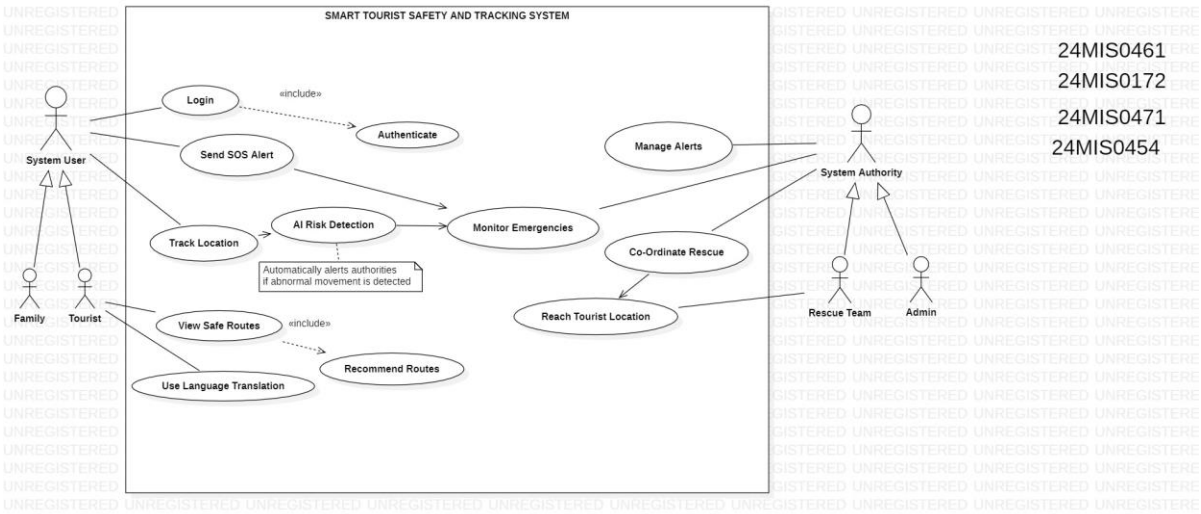
If any risky situation is detected, such as no movement or unusual movement, the system automatically generates an alert.

The alert is sent to the admin and family members for quick response.

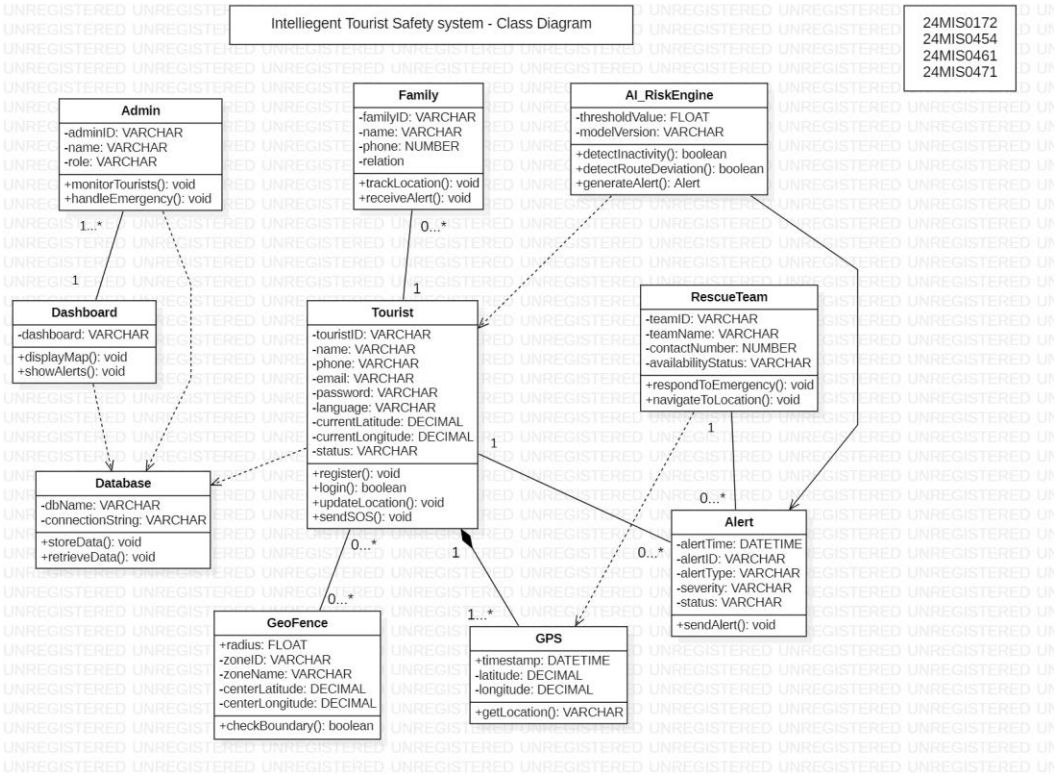
Finally, when the trip ends, the tracking process is stopped.

4. UML Diagrams

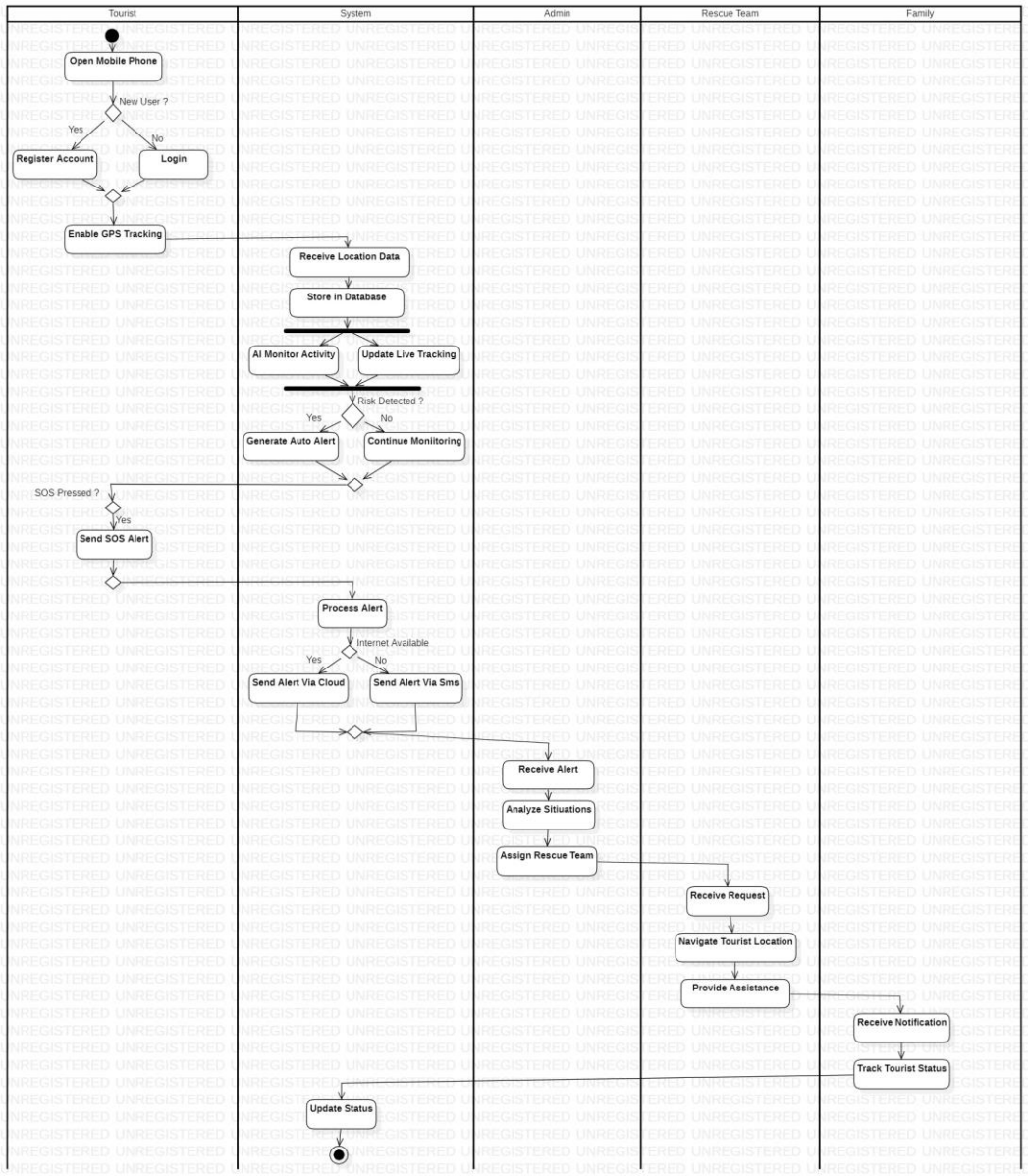
4.1 Use Case Diagram



4.2 Class Diagram

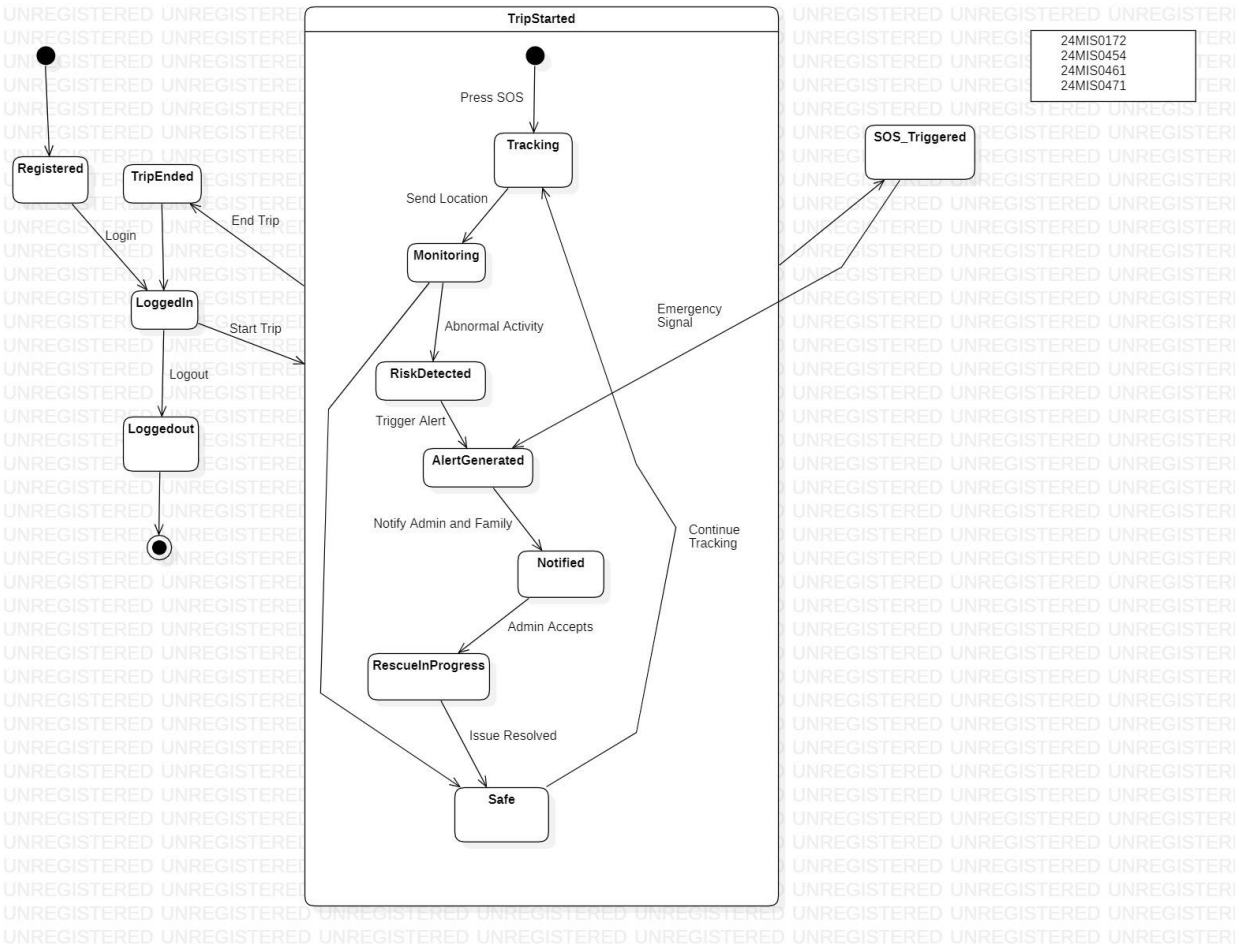


4.3 Activity Diagram



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4.4 State Diagram



5. Implementation

The Smart Tourist Safety and Tracking System is designed using a modular approach, focusing on simplicity and usability.

The system includes modules such as user management, location tracking, alert system, and monitoring.

The location tracking module continuously updates the tourist's position using GPS. The data is sent to the backend for processing.

The alert system is designed to generate alerts automatically when abnormal conditions are detected. These alerts are sent to admin and family members.

The backend handles all processing and communication between modules, while the database stores all required data.

The frontend provides a simple dashboard where users can view tracking information and alerts.

The system is tested using sample scenarios to check how it works in normal and emergency conditions. The results show that the system can respond quickly and provide useful information.

Version control is maintained using GitHub during the design process.

6. Results and Analysis

The proposed Smart Tourist Safety and Tracking System was analyzed using different test scenarios and expected use cases. The system was checked based on how well it can track location, detect risk situations, and send alerts.

The tracking module was able to provide continuous location updates, which helps in knowing the exact position of the tourist. This makes it easier to monitor the movement during the trip.

The alert system showed quick response when abnormal situations were considered, such as no movement for a long time or sudden change in location. Alerts were generated automatically and can be sent to admin and family members.

Compared to existing systems, the proposed system reduces delay because it does not depend only on manual actions. It works automatically and provides faster response.

The system also improves communication between users. Admin can easily monitor multiple tourists, and family members can get updates during emergency situations.

Overall, the system shows better performance in terms of safety, response time, and coordination. It is simple, practical, and suitable for real-world usage if implemented.

6.1 Comparison Table

FEATURE	EXISTING SYSTEM	PROPOSED SYSTEM
Tracking	Not Available	Real-Time GPS
Alerts	Manual	Automatic+Manual
Response Time	Slow	Fast
Coordination	Poor	Integrated
Safety Level	Low	High

6.2 Performance Improvement Table

PARAMETER	BEFORE	AFTER
Response Time	High	Low
Accuracy	Medium	High
Reliability	Low	High

7. Conclusion and Future Work

Conclusion

The Smart Tourist Safety and Tracking System designed in this project provides a simple and effective solution to improve tourist safety. By using tracking and alert features, the system helps in handling emergency situations more quickly.

The system uses a structured design approach and combines different modules like tracking, monitoring, and alert system. This makes it easy to understand and useful for real-world application.

The results show that the system can reduce delay, improve communication, and provide better safety support for tourists.

Future Work

Even though the system design is effective, there are some areas that can be improved in future.

- The system can be developed as a real mobile application
- More accurate AI models can be used for better risk detection
- Additional features like voice alerts can be added
- The system can be connected with police or emergency services
- Wearable devices can be used for better tracking

These improvements can make the system more powerful and useful.

8. Team Contribution

- **Madhan Kumaran R:** Worked on system idea, design, and overall planning
- **Sanjay Guptha R S:** Designed class diagram and helped in system structure
- **Thirukumaran K:** Worked on activity diagram and workflow design
- **Abdullah S:** Prepared documentation and collected data

All team members contributed equally and worked together to complete the project successfully.

9. References

- OOAD class notes .
- UML design materials .
- Online articles related to tourist safety.
- Research papers on tracking systems .

REVIEW 1 PPT LINK:

OOAD COURSE PROJECT PPT 1

REVIEW 2 PPT LINK:

OOAD COURSE PROJECT PPT2